PEER REVIEW HISTORY

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ARTICLE DETAILS

TITLE (PROVISIONAL)	Efficacy and safety of anti-epidermal growth factor receptor agents for the treatment of oesophageal cancer: a systematic review and meta-analysis
AUTHORS	Zhang, Lijuan; Song, Yanli; Jiang, Nan; Huang, Yaqi; Dong, Bo; Li, Wei; He, Yanze; Chen, Yun; Liu, Haibin; Yu, Rui

VERSION 1 – REVIEW

REVIEWER	Amanda Lee University of Hull Faculty of Health Sciences, UK
REVIEW RETURNED	23-Nov-2020

GENERAL COMMENTS	A very interesting paper - I do hope this feedback is useful. Abstract Please identify in your abstract whether you followed any guidelines for openness and transparency in systematic review processes. (PRISMA) The abstract does not meet criteria for a systematic review paper… For example, you cite participants as patients with OC – but data were collected from RCTs … The nature of Meta analysis would mean your inclusion criteria was : all RCTs on treatments for OC with or without anti EGFR… where data could be extracted for each arm of the study. Introduction – how do authors define parameters of OC? (ICD C15/16?) Can you highlight to readers the extent of 'conventional therapies' which were included? How do you factor in the confounding variables on survival outcomes across your papers? (ie increasing age & demographics / histology / presentation stage)? I see you used the outcome measures of OS and/or PFS – so can you make this clear to the readers? It would be helpful to identify the biological plausibility of EGFR in reduction of symptoms in OC, to define OC accordingly and to identify geographic disparities in OC presentation, survival and clinical outcomes within the introduction / background. Search strategy - ? terms to capture *esophagus / gastroesophageal / *esophagogastric / - Perhaps the authors could just add information to support how they defined oesophageal cancer in light of no mention of ICD classification? (ICD C15.0 to C15.9). PRISMA – how did authors separate studies focusing on OC only (many studies focus on GOC with or without gastric / or even upper GI / head and neck infiltration) – how did authors define OC for this Meta analysis?

	TABLE S1 – there are some historical studies… How do authors control for increasing survival rates in more recent years (with advancing therapies and treatments available)? (ie neoadjuvant therapies introduced more recently)? How did authors control for stage at presentation and decreased survival ? A significant number of single institution studies across China – this could be a limitation? What control measures were taken to ensure there were no affiliations with drug companies across trials?

REVIEWER	Dr Mohammad H Derakhshan University of Glasgow, UK
REVIEW RETURNED	27-Nov-2020

GENERAL COMMENTS	I enjoyed reading the manuscript. In terms of statistical methods, the authors have employed robust techniques for systematic review and meta-analysis. This includes methods for search strategy, primary and secondary screening, data extraction, assessment of quality, assessment of bias, pool estimates, forest plots, and sensitivity analyses. However, to improve the presentation and attract more confidence in the results I would suggest: 1. To assess the publication bias I see authors have prepared a funnel chart (fig 4) for the whole study. I recommend having one chart for each outcome separately if the number of studies is 10 or more, i.e. for OS, ORR, and DCR. 2. To clarify the mismatch between Table S1 and flowchart regarding the final number of included studies (24 vs 25). 3. To clarify whether the five excludes studies (page 9 line 18) are considered in the final 25 studies the flow chart.

VERSION 1 – AUTHOR RESPONSE

Reviewer: 1
Dr. Amanda Lee, University of Hull

Comments to the Author
A very interesting paper - I do hope this feedback is useful.

Abstract
1. Please identify in your abstract whether you followed any guidelines for openness and transparency in systematic review processes. (PRISMA)
Response: Thanks for your suggestion. We have added a sentence marked with yellow in the part of Abstract.

Abstract
Results: We followed the Preferred Reporting Items for Systematic reviews and Meta-Analysis (PRISMA) guidelines.

2. The abstract does not meet criteria for a systematic review paper… For example, you cite participants as patients with OC – but data were collected from RCTs … The nature of Meta analysis would mean your inclusion criteria was :
all RCTs on treatments for OC with or without anti EGFR… where data could be extracted for each arm of the study.
Response: Thanks for your comments. The PICOS of this systematic review and meta-analysis was as followed. We have revised the inappropriate statements in the abstract and then marked them with yellow.
Patient: Patients with OC
Intervention: Anti-EGFR treatment
Comparison: Conventional treatment (other treatment modalities or drugs)
Outcome: Overall survival (OS), Progression-free survival (PFS), Objective response rate (ORR), Disease control rate (DCR), and grade 3-5 toxicities (including haematological toxicities and non-haematological toxicities)
Study design: Randomised controlled trials (RCTs).

Abstract
Participants: Patients with OC.
Interventions: We searched PubMed, EMBASE, Web of Science, ClinicalTrials.gov, Cochrane Library, Chinese Biology Medicine, China National Knowledge Infrastructure, and Wanfang Data Knowledge Service Platform for RCTs of anti-EGFR treatment versus conventional treatment in patients with OC, from inception to December 2019.

3. Introduction – how do authors define parameters of OC? (ICD C15/16?) Can you highlight to readers the extent of 'conventional therapies' which were included?
Response: Thanks for your good question and suggestion. In this systematic review and meta-analysis, we could not define parameters of esophageal cancer (OC) in the part of the introduction because of the included studies did not provide the coding of the International Classification of Diseases (ICD) for readers. Therefore, patients with histologically confirmed OC described in original articles were included for our meta-analysis. The definition of OC was consistent with the guidelines of the eighth edition staging for OC.1 We have revised sentences in the part of the Inclusion and Exclusion Criteria and added a sentence in the part of the Literature Search Strategy as followed. Please see the yellow text.
In our paper, conventional therapies referred to any single-modality or multimodality treatments for OC, such as surgery, chemotherapy (CT), radiotherapy (RT), or any combinations of them, which were mentioned in the part of the Introduction. Compared with anti-epidermal growth factor receptor (anti-EGFR) treatment for OC, all of those mentioned above were named conventional therapies. We have made an explanation as followed and highlighted the changes by yellow text in the part of the Introduction and the Inclusion and Exclusion Criteria.
Reference
1 Rice TW, Ishwaran H, Ferguson MK, et al. Cancer of the Esophagus and Esophagogastric Junction: An Eighth Edition Staging Primer. J Thorac Oncol 2017; 12: 36-42.

Inclusion and Exclusion Criteria
Eligible studies were included if they met the following criteria: (i) patients with histologically confirmed OC;
Literature Search Strategy
Location of OC primary site, including typical endoscopic measurements of each region measured from the incisors (range from 15cm to 42cm), which is defined as beginning from the upper oesophageal sphincter to the proximal 2 cm of the cardia.23
Introduction
Therefore, the addition of novel therapies to conventional treatments (referred to any single-modality

or multimodality therapies were mentioned above) is urgently needed.
Inclusion and Exclusion Criteria
(2) anti-EGFR treatment versus conventional treatment (referred to any single-modality or multimodality treatments for OC, such as surgery, CT, RT, or any combinations of them);
4. How do you factor in the confounding variables on survival outcomes across your papers? (ie increasing age & demographics / histology / presentation stage)? I see you used the outcome measures of OS and/or PFS – so can you make this clear to the readers?
Response: Thanks for your good question. Subgroup analysis was performed to evaluate whether confounding variables influenced the survival outcomes as you mentioned OS (Figure S1) and/or PFS (Figure S2). However, the outcome measures of OS and/or PFS were calculated with the data of the Kaplan-Meier curves that made it impossible to extract the corresponding data for some confounding variables (ie increasing age & demographics/histology/presentation stage). We have revised the original sentence into “Both the data of the OS and PFS were got from Kaplan–Meier curves, which were read using Engauge Digitizer version 10.8 (free software downloaded from http://sourceforge.net/projects/digitizer/).” in the Statistical Analysis Part. Please see the yellow text.

Statistical Analysis
Both the data of the OS and PFS were got from Kaplan–Meier curves, which were read using Engauge Digitizer version 10.8 (free software downloaded from http://sourceforge.net/projects/digitizer/).

5. It would be helpful to identify the biological plausibility of EGFR in reduction of symptoms in OC, to define OC accordingly and to identify geographic disparities in OC presentation, survival and clinical outcomes within the introduction / background.
Response: Thank you very much for your constructive suggestions. For the first question, we have added sentences in the part of the Introduction and marked with yellow.
The definition of OC has been described in the part of the Literature Search Strategy and marked with yellow.
We have demonstrated the geographic disparities in OC presentation, survival and clinical outcomes within the Introduction.

Introduction
Tyrosine kinase inhibitors (TKIs) and monoclonal antibodies (mAbs) are two promising anti-EGFR drugs. mAbs bind to the extracellular domain of EGFR and compete with endogenous ligands to block the ligand-induced EGFR tyrosine kinase activation by blocking the ligand-binding region.13 TKIs compete reversibly with adenosine 5' triphosphate to bind to the intracellular catalytic domain of EGFR tyrosine kinase and inhibit the EGFR autophosphorylation and downstream signaling.13
Literature Search Strategy
Location of OC primary site, including typical endoscopic measurements of each region measured from the incisors (range from 15cm to 42cm), which is defined as beginning from the upper oesophageal sphincter to the proximal 2 cm of the cardia.23
Introduction
However, its incidence and mortality rates vary considerably, as demonstrated by a two- to three-fold difference in these statistics between regions all over the world.2 OC is common in Eastern and Southern Africa, and the highest incidence rates have been reported in East Asia, with rates in Mongolia and China being among the top five worldwide.1 2 The highest-risk area, called “oesophageal cancer belt”, where squamous cell cancers dominated all cases by 90%, and adenocarcinoma is largely a disease of white.1
OC is diagnosed in the later stages in 80% of the patients. Typically, few patients benefit from single-modality therapies, such as surgery [(15%-20% rates of 5-year (overall survival, OS)], chemotherapy (CT, 15%-45% rates of 5-year OS), and radiotherapy (RT, 0%-15% rates of 5-year OS).3 Therefore, chemoradiotherapy (CRT, 26% rates of 5-year overall survival) has become a standard of care for

OC.

6. Search strategy - ? terms to capture *esophagus / gastroesophageal / *esophagogastric / - Perhaps the authors could just add information to support how they defined oesophageal cancer in light of no mention of ICD classification? (ICD C15.0 to C15.9).
Response: Thanks for your suggestion. We have added information to support how we defined OC in this systematic review and meta-analysis. Please see the yellow text in the part of the Literature Search Strategy.

Literature Search Strategy
Location of OC primary site, including typical endoscopic measurements of each region measured from the incisors (range from 15cm to 42cm), which is defined as beginning from the upper oesophageal sphincter to the proximal 2 cm of the cardia.23

7. PRISMA – how did authors separate studies focusing on OC only (many studies focus on GOC with or without gastric / or even upper GI / head and neck infiltration)
– how did authors define OC for this Meta analysis?
Response: Thanks for your good questions. According to the guidelines of the eighth edition staging for OC,1 patients with histologically confirmed OC were described in original articles would be included for our meta-analysis. However, studies focused on GOC with or without gastric/or even upper GI/head and neck infiltration were not in compliance with the guidelines of the eighth edition staging for OC, which were excluded in our meta-analysis. Furthermore, the process of screening the studies based on the selection criteria by two trained researchers, and any disagreements were settled by discussion and/or consultation with other researchers and/or clinical experts.
In this meta-analysis, the OC is defined as followed. We have added the following yellow text in the part of the Literature Search Strategy.
Reference
1 Rice TW, Ishwaran H, Ferguson MK, et al. Cancer of the Esophagus and Esophagogastric Junction: An Eighth Edition Staging Primer. J Thorac Oncol 2017; 12: 36-42.

Literature Search Strategy
Location of OC primary site, including typical endoscopic measurements of each region measured from the incisors (range from 15cm to 42cm), which is defined as beginning from the upper oesophageal sphincter to the proximal 2 cm of the cardia.23

8. TABLE S1 – there are some historical studies… How do authors control for increasing survival rates in more recent years (with advancing therapies and treatments available)? (ie neoadjuvant therapies introduced more recently)?
Response: Thank you for your question. With the improving diagnostic techniques (i.e., the multidetector computed tomography, the wider availability of positron emission tomography scan, endoscopic ultrasonography and bronchoscopy) and advancing therapies and treatments (multimodality strategies, such as surgery, chemotherapy, radiotherapy, concurrent chemoradiotherapy, target agents therapy, or any combinations of them for different kinds of patients), the prognosis of OC got better and the survival rates increased in more recent years.

9. How did authors control for stage at presentation and decreased survival?
Response: Thanks for your question. OC is one of the most common cancers worldwide. Survival rates are very low and advanced OC carries an overall poor prognosis with most patients presenting with incurable disease. The disease is highly lethal, with overall 5-year survival rates of only 10-15%.1 Frequently the diagnosis is made when the patients present with dysphagia due to an obstructing tumour.2 The disease is then already in an advanced stage with a high potential of occult metastases, which makes the survival decreased. Over the years diagnostic techniques have improved, the

multidetector computed tomography and the wider availability of PET-scan made better staging of the tumour.2 This improved staging may have resulted in stage migration and better stage-specific survival. However, this phenomenon generally does not improve survival of all stages combined.
Reference
1 Enzinger PC, Mayer RJ. Esophageal cancer. N Engl J Med 2003; 349: 2241-52.
2 Crane LM, Schaapveld M, Visser O, et al. Oesophageal cancer in The Netherlands: increasing incidence and mortality but improving survival. Eur J Cancer 2007; 43: 1445-51.

10. A significant number of single institution studies across China – this could be a limitation? What control measures were taken to ensure there were no affiliations with drug companies across trials?
Response: Thanks for your question. A significant number of single institution studies across China with small sample size might be a limitation, which made the representativeness of samples was poor. It has been demonstrated as a limitation in the part of the discussion. Further high quality large-scale and multi-centre studies are needed to confirm the efficacy and safety of anti-EGFR agents in OC.
The second question was not involving in our systematic review and meta-analysis. The included studies did not report the drug companies involving in their study design. Therefore, no other control measures were taken to ensure affiliations with drug companies across trials.

Discussion
Third, given the geographical variation incidence rates of OC, most of the included studies were conducted in China with small sample size in single institutions. Therefore, physicians from other countries should consider carefully when apply anti-EGFR agents in their clinical practices.

Reviewer: 2
Dr. Mohammad Derakhshan, University of Glasgow

Comments to the Author
I enjoyed reading the manuscript. In terms of statistical methods, the authors have employed robust techniques for systematic review and meta-analysis. This includes methods for search strategy, primary and secondary screening, data extraction, assessment of quality, assessment of bias, pool estimates, forest plots, and sensitivity analyses. However, to improve the presentation and attract more confidence in the results I would suggest:

1. To assess the publication bias I see authors have prepared a funnel chart (fig 4) for the whole study. I recommend having one chart for each outcome separately if the number of studies is 10 or more, i.e. for OS, ORR, and DCR.
Response: Thanks for your good suggestion. We have added two funnel charts in the part of the Publication Bias. Please see the Figure S12 for ORR and Figure S13 for DCR. Figure 4 is the funnel plot of the OS.

Figure S12 Funnel plot of publication bias for ORR

Figure S13 Funnel plot of publication bias for DCR

2. To clarify the mismatch between Table S1 and flowchart regarding the final number of included studies (24 vs 25).
Response: Thanks for your suggestion. We have checked the match between the Table S1 and the flowchart (Figure 1) regarding the final number of the included studies again. There are 25 studies in Table S1, which are consistent with the flowchart. Two studies from the UK were performed on the same group of study patients and had partly overlapping study periods. However, different outcomes

were analysed in each study. Therefore, both of the two studies were included in this systematic review and meta-analysis. We have described it in the part of the Study Characteristics and marked with yellow.

Study Characteristics
Two studies from the UK were performed on the same group of study patients and had partly overlapping study periods.40 41 However, different outcomes were analysed in each study.

3. To clarify whether the five excludes studies (page 9 line 18) are considered in the final 25 studies the flow chart.
Response: Thanks for your suggestion. Five excluded studies are not considered in the final 25 studies the flow chart.

VERSION 2 – REVIEW

REVIEWER	Amanda Lee University of Hull Faculty of Health Sciences
REVIEW RETURNED	09-Feb-2021

GENERAL COMMENTS	I have read the amended submission and can see authors have addressed comments I made at initial peer review.

REVIEWER	Dr Mohammad H Derakhshan University of Glasgow
REVIEW RETURNED	05-Feb-2021

GENERAL COMMENTS	I am satisfied with the revisions in the new version. No further changes are required.

VERSION 2 – AUTHOR RESPONSE

Reviewer: 1
Dr. Amanda Lee, University of Hull
Comments to the Author:
I have read the amended submission and can see authors have addressed comments I made at initial peer review.
Response: Thank you very much.

Reviewer: 2
Dr. Mohammad Derakhshan, University of Glasgow
Comments to the Author:
I am satisfied with the revisions in the new version. No further changes are required.
Response: Thank you very much.